

Flood Disaster Response System

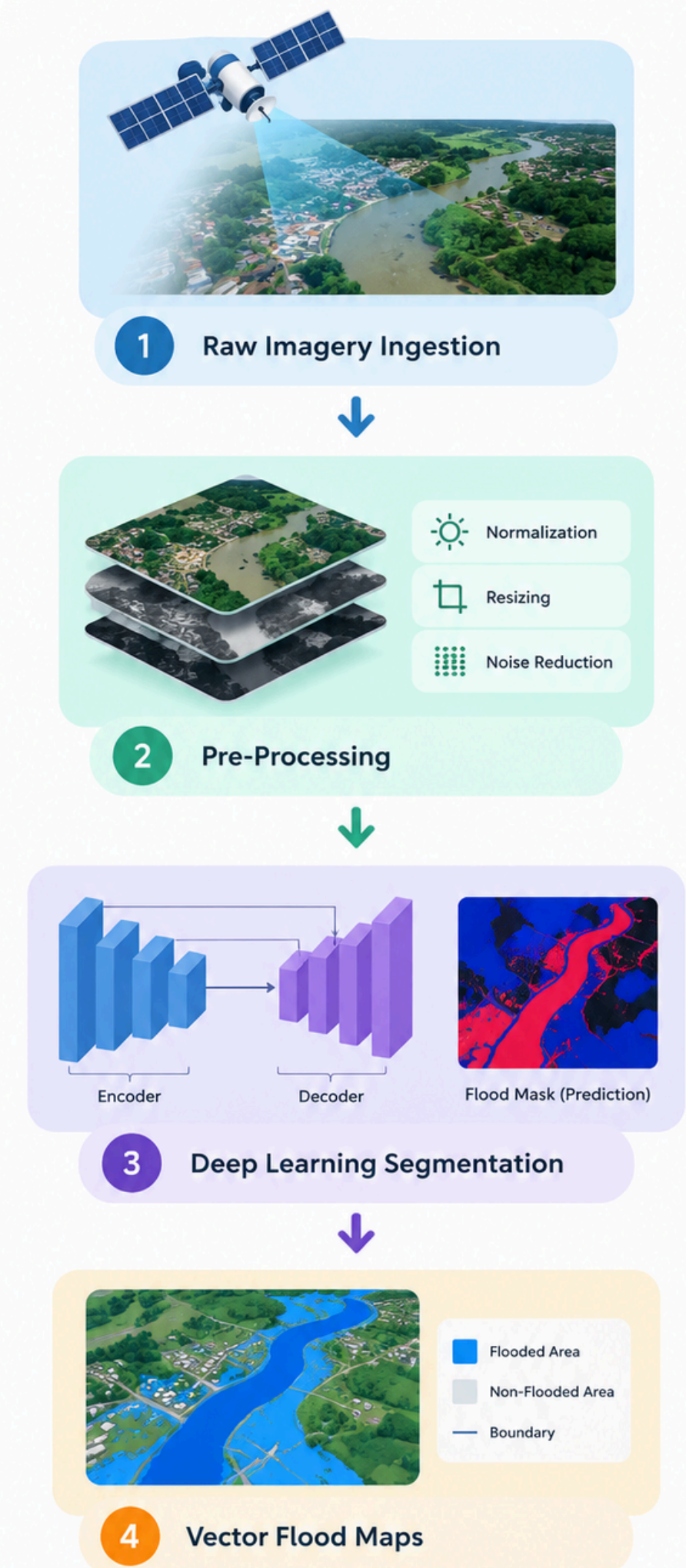
Computer vision for pixel-level flood segmentation using U-Net

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1. Problem & Objective

- Task: Classify every pixel as flooded or non-flooded
- Input: RGB flood image
- Output: Binary pixel-level flood mask
- Why Pixel-Level Segmentation?
 - Flooding has irregular boundaries
 - Identifies the exact location and extent of flooded areas
 - Provides more detailed information than simply classifying an entire image as flooded or non-flooded



2. Data & Preprocessing

- Dataset: Kaggle Flood Area Segmentation
- Total Data: 283 paired images and masks
- Data Split: 198 training images
- 42 validation images
- 43 test images
- Image Preprocessing: Resize images to 256 × 256 pixels
- Apply ImageNet mean/std normalization
- Data Augmentation: Horizontal flip
- Vertical flip
- Brightness and contrast adjustment
- Reproducibility: A fixed split.csv is used to maintain the same train, validation, and test assignments

3. U-Net Architecture & Why It Was Chosen



MODEL

- U-Net with ResNet34 encoder
- ImageNet-pretrained weights

TRAINING

- 30 epochs
- Batch size: 8
- Optimizer: Adam
- Learning rate: 3×10^{-4}
- Loss: BCE + Dice

WHY U-NET?

- Encoder learns visual features
- Decoder reconstructs pixel-level masks
- Skip connections preserve spatial details
- Suitable for irregular flood boundaries

4. Validation Results

IoU

Best validation IoU: 82.7%

Precision

Best validation precision:
89.5%

Recall

Best validation recall:
91.5%

F1-score

Best validation Dice: 90.5%

Best epoch: 21 / 30

IoU measures overlap between predicted and true flood pixels. Higher IoU and Dice indicate better mask agreement.

Validation metrics are strong and balanced. These are validation results, not aggregate test-set results.



5. End-to-End Computer Vision Pipeline

The trained model converts an RGB image into a binary flood mask through five reproducible stages.

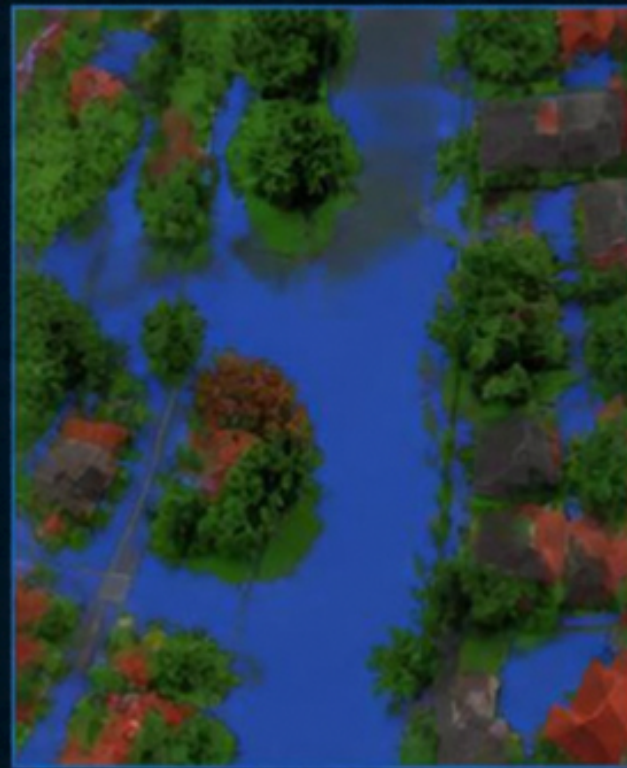
Input\nRGB
flood image



Preprocess\n256x256 +
normalize



U-
Net\nResNet
34 encoder



Prediction
\n0.5
threshold



Output\nBinary
flood mask

